

Mining Weighted Frequent Itemsets

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Abstract

To enhance the effectiveness of reducing unpromising candidates in weighted frequent itemset mining, this work presents an improved model to tighten more accurate upper-bounds of weighted supports for itemsets in mining. In addition, a projection-based mining approach is proposed to efficiently find weighted frequent itemsets. The experimental results show that the proposed approach has good execution efficiency.

1 Introduction

Data mining techniques can be used to extract useful rules or patterns from different kinds of databases. Among data mining techniques, association-rule mining [1][2] is one of important issues due to the consideration of frequency relationship of items. For example, in retailing business, the two products, milks and breads, are frequently sold in transactions. In association-rule mining, however, although the frequency relationship of items in transactions is considered, the same importance is assumed for all items in databases. Thus, the actual significance of an itemset cannot be easily recognized by the association-rule mining techniques. For example, the significance of the two products, DVD Players and milks, in a store is not the same according to their individual profits.

To address this problem, Cai *et al.* first presented a new issue named weighted frequent itemset mining [3] that different weight values were assumed for items according to the importance of the items. In Cai *et al.*'s study [3],

the weighted support of an itemset was the support value of the item multiplied by the summation of weight values of items in the itemset. Based on the function, the weight for an itemset is still not easily obtained.

Due to this reason, Yun *et al.* presented an average-weight concept [6], which was the sum of weights of items in an itemset divided by the number of items in that itemset, to evaluate the individual weight of an itemset. In addition, the different weights of an item in different conditions were permitted. Based on this function, the downward-closure property was not satisfied, so an upper-bound model was developed to avoid information loss in any weighed frequent itemset mining case. The main concept of the model [6] is that the maximum value of the weights of all items in a database is used as the weight upper-bound value of each transaction to construct the downward-closure property. However, since not all transactions in a database include the items with maximum weight values for all items, a large number of unpromising candidate itemsets in the mining process are generated by the model [6]. To efficiently enhance the problem of mining weighted frequent itemsets, developing a proper upper-bound model to find weighted frequent itemsets is thus critical.

To address the above reason, the work presents an effective model to efficiently find weighted frequent itemsets in transactions. In addition, a new improved upper-bound model named transaction-weight upper-bound model is developed to obtain more accurate upper-bounds of weighted values for itemsets in mining. Based on this model, the projection-based mining

approach is developed to efficiently handle the problem of mining weighted frequent itemsets. The experimental results show the proposed algorithm achieves good performance in execution efficiency.

The rest of this paper is organized as follows. Some related works are briefly reviewed in Section 2. The problem statement and related definitions are given in Section 3. The proposed *PWA* algorithm is described in Section 4. The experimental evaluation is shown in Section 5. Conclusions and suggestions for future work are given in Section 6.

2 Related Works

In the field of data mining, one common type of data mining is to derive association rules from a transaction dataset, such that the presence of certain item in a transaction will imply the presence of some other items. However, although traditional association-rule mining considers the common occurrence relationship of items in a database, the same significance is assumed for all the items in that database. In reality, the importance of items in a database may vary depending on various factors, such as profit and cost. For example, if the significance values of products in a retailing store are considered by the profits of the items, then their significance values are different, such as the profit of LCD TVs is different from that of milks, and its profit is much higher than another. Thus, some items with high-important may not be discovered by using traditional frequent itemset mining techniques.

To address this, Cai *et al.* first proposed a new research topic, namely weighted frequent itemset mining [3], which considered not only the occurrence of items in a set of transactions but also the individual weight values of items, to find meaning rules in databases. Based on this function [3], however, the downward-closure property in association-rule mining cannot be kept in the weighted frequent itemset mining problem. Hence, an effective k -boundary model was developed to satisfy the downward-closure property and find all weighed frequent itemsets from databases.

Afterward, Yun *et al.* [6] proposed a new average-weight function to evaluate the weight value of an itemset, which was the sum of weight values of items in an itemset divided by the number of items in that itemset [6]. Through the function, the significance for each itemset in a database could be recognized.

However, there exists a challenge in weighted frequent itemset mining. That is, the downward-closure property in association-rule mining cannot be maintained with the

average-weight function. To solve this problem, Yun *et al.* proposed an upper-bound model [6] to keep the downward-closure property in weighted itemset mining. The main concept of the model is that the maximum weight value among all items in a database was regarded as the upper-bound of the weight value of each transaction in that database, and then the upper-bound of a transaction could be used as the upper-bounds of weight values for itemsets in that transaction.

However, although the model [6] can be used to reduce the search space of finding weighted frequent itemsets in comparison with a complete search space (e.g. a lattice), not all transactions included the items with the maximum weight value. Thus, the upper-bounds of weight values of itemsets are overly overestimated by the model, and also a large number of unpromising itemsets are generated in mining. Accordingly, how to effectively reduce the number of unpromising itemsets in weighted frequent itemset mining is critical.

3 Problem Statement and Definitions

In this section, to clearly describe the problem of weighted frequent itemset mining, a transaction database is given in Table 1, in which each transaction consists of two features, the transaction identification (*TID*) and items purchased. There are eight items in the transactions, respectively denoted as *A* to *H*. The predefined weight values of the eight items are 0.30, 0.60, 0.45, 0.20, 0.40, 0.50, 0.15 and 0.95.

Table 1: The set of the five transactions.

<i>TID</i>	<i>Transactions</i>
<i>Trans</i> ₁	{ <i>ACDEF</i> }
<i>Trans</i> ₂	{ <i>BH</i> }
<i>Trans</i> ₃	{ <i>BCE</i> }
<i>Trans</i> ₄	{ <i>ADEGH</i> }
<i>Trans</i> ₅	{ <i>ADEF</i> }

In our study, an itemset X is a subset of items, $X \subseteq I$. If $|X| = r$, the itemset X is called an r -itemset. Here $I = \{i_1, i_2, \dots, i_m\}$ is a set of items, which may appear in transactions. A transaction database TDB is composed of a set of transactions. That is, $TDB = \{Trans_1, Trans_2, \dots, Trans_y, \dots, Trans_z\}$, where $Trans_y$ is the y -th transaction in TDB . Here the predefined weight of an item i , w_i , is in the range from 0 to 1. For example, $w_A = 0.30$.

For an itemset X , the weight (w_X) of X in a transaction is the sum of the weights of all items in X divided by the number of items in X in a transaction. For example, the weights of the two items in the itemset $\{AD\}$ are 0.3 and 0.2,

respectively, and the total number of items in $\{AD\}$ is 2. Hence, $w_{\{AD\}} = (0.30 + 0.2) / 2 = 0.25$. Furthermore, the weighted support (ws_X) of an itemset X in a transaction database TDB is the sum of the weights of X in transactions including X in TDB . For example, the weights of the two items in the itemset $\{AD\}$ are 0.3 and 0.2, respectively, and the total number of items in $\{AD\}$ is 2. Hence, $w_{\{AD\}} = (0.30 + 0.2) / 2 = 0.25$. Finally, let λ be a pre-defined minimum weighted support threshold. An itemset X is called a weighted frequent itemset (abbreviated as WF) if $wsup_X \geq \lambda$. For example in Table 1, if $\lambda = 0.7$, then $\{AD\}$ is a weighted frequent itemset since $wsup_{\{AD\}} = 0.75 \geq \lambda$.

However, the downward-closure property in traditional frequent itemset mining is not kept in weighted itemset mining. For example, in Table 1, item D appears in three transactions $Trans_1$, $Trans_4$ and $Trans_5$, and its weight value is 0.2. Since the weighted support value of $\{D\}$ is 0.6 ($= 0.2 + 0.2 + 0.2$), $\{D\}$ is not a weighted frequent itemset. However, its super-itemset $\{AD\}$ is a weighted frequent itemset. As this example notes, the information losing is possibly appeared in weighted frequent itemset mining problem.

In this study, hence, an effective upper-bound model can be developed in the weighted frequent itemset mining problem. The main concept is that the maximum weight value among items in a transaction is used as the upper-bounds of weights of that transaction. For example, in Table 1, the third transaction $Trans_5$ only includes the four items, A , D , E and F , and their weight values are 0.30, 0.20, 0.40 and 0.50, respectively. Since the maximum weight value among the four items is the weight value ($= 0.5$) of item F , the transaction-weight upper-bound $twub_5$ of $Trans_5$ is 0.5. Then, the weighted support upper-bound of an itemset is the summation of transaction-weight upper-bounds of the transactions including the itemset in the database. For example, in tables 1 and 2, the itemset $\{AD\}$ appears in the first, fourth and fifth transactions, and the transaction-weight upper-bounds of the three transactions are 0.50, 0.95 and 0.50, respectively. Then, the weighted support upper-bound value of $\{AD\}$ can be calculated as $0.50 + 0.95 + 0.50$, which is 1.95. At last, an itemset X is called a weighted frequent upper-bound itemset ($WFUB$) if $wsup_X \geq \lambda$. For example, if $\lambda = 0.7$, then the itemset $\{AD\}$ is a weighted frequent upper-bound itemset.

Based on the definitions above, a weighted itemset considers not only the occurrence of items in transactions but also the individual weights of the items. The problem to be solved in this work is to find all weighted frequent itemsets, whose weighted supports are larger than or equal to a pre-defined minimum weighted support threshold

λ from a given transaction database, TDB . The details of the proposed upper-bound model and algorithm are then stated in the next section.

4 Proposed Algorithm

In this section, the improved model is first described below.

1. The Improved Upper-bound Model

As mentioned previously, although the existing upper-bound model [6], can be used to avoid information losing in any weighted frequent itemset mining case, the model [6] adopts the maximum weight value among all items in a database as the upper-bound of weight value of each transaction in that database. However, not all transactions include the items with the maximum weight value in that database, and then a large number of unpromising candidate itemsets are generated by using the traditional model. Accordingly, the work presents an improved model that flexibly adopts the maximum weight value among all items in a transaction as the upper-bound of that transaction. The developed model can effectively be applied to tighten upper-bounds of weighted support values for itemsets in mining when compared to the existing model, and thus speeding up the execution efficiency in terms of finding weighted frequent itemsets.

2. The Proposed Mining algorithm, PWA

The execution procedure of the proposed projection-based weighted itemset mining algorithm (named PWA) is then stated below.

INPUT: A set of items, each with a predefined weight value; a transaction database TDB , in which each transaction includes a subset of sold items; a predefined minimum weighted support threshold λ .

OUTPUT: A final set of weighted frequent itemsets, WFs .

STEP 1: For each y -th transaction $Trans_y$ in TDB , find the transaction-weight upper-bound $twub_y$ of $Trans_y$ as:

$$twub_y = \max\{w_{y1}, w_{y2}, \dots, w_{yj}\},$$

where w_{yj} is the weight w_i of each j -th item i in $Trans_y$.

STEP 2: For each item i in TDB , do the following substeps.

- (a) Find the weighted support count value $wsup_i$ of the item i as:

$$ws_{ub_i} = \sum_{i \subseteq Trans_y \wedge Trans_y \subseteq TDB} twub_y.$$

- (b) Find the weighted support ws_i of the item i as:

$$ws_i = \frac{\sum_{i \subseteq Trans_y \wedge Trans_y \subseteq TDB} w_{yi}}{|i|}.$$

STEP 3: For each item i in TDB , do the following substeps.

- (a) If the weighted support upper-bound ws_{ub_i} of i is larger than or equal to the minimum weighted support threshold λ , put it in the set of weighted frequent upper-bound I -itemsets, $WFUB_I$.
- (b) If the weighted support ws_i of i is larger than or equal to the minimum weighted support threshold λ , put it in the set of weighted frequent I -itemsets, WF_I .

STEP 4: Process each itemset X in the set of $WFUB_I$ in alphabetical order by the following substeps.

- (a) Find the relevant transactions including X in TDB , and put the transactions in the set of projected transactions tdb_X of X .
- (b) Find all the weighted frequent itemsets with X as their prefix itemsets by the *Finding-WF*(X, tdb_X, I) procedure. Let the set of returned weighted frequent itemsets be WF_X .

STEP 5: Output the final set of weighted frequent itemsets in all the WF_X .

After STEP 5, all the weighted frequent itemsets are found by *Finding-WF*(x, tdb_x, r) procedure, and the procedure is stated below.

The *Finding-WF*(X, tdb_X, r) procedure:

Input: A prefix r -itemset X and its corresponding projected transactions tdb_X .

Output: The weighted frequent itemsets with the prefix itemset X .

PSTEP 1: Initialize the temporary itemset TI_X table as an empty table, in which each tuple consists of three fields: itemset, weighted support upper-bound ($twub$) of the transaction, and the weighted support (ws) of the itemset.

PSTEP 2: For each y -th transaction $Trans_y$ in tdb_X , do the following substeps.

- (a) Get each item i located after X in $Trans_y$.
- (b) Generate the $(r+1)$ -itemset X' consisting of the prefix r -itemset X and i , put the itemset X' , the transaction-weight upper-bound of $Trans_y$ and the actual weight value of X'

in the TI_X table.

PSTEP 3: For each $(r+1)$ -itemset X' in the TI_X table, do the following substeps.

- (a) If the weighted support upper-bound $ws_{ub_{X'}}$ of X' is larger than or equal to the minimum weighted support threshold λ , put it in the set of weighted frequent upper-bound $(r+1)$ -itemsets with the prefix itemset X , $WFUB_{(r+1), X}$.
- (b) If the actual weighted support $ws_{X'}$ of X' is larger than or equal to the minimum weighted support threshold λ , put it in the set of weighted frequent itemsets, $WF_{(r+1), X}$.

PSTEP 4: Process each $(r+1)$ -itemset X' in $WFUB_{r+1, X}$ by the following substeps.

- (a) Find the relevant transactions including X' from tdb_X , and then put the transactions including X' in the set of projected transactions $tdb_{X'}$ of X' .
- (b) Find all weighted frequent itemsets with X' as their prefix itemsets by the *Finding-WF*($X', tdb_{X'}, r+1$) procedure, where $x = X'$ and $tdb_x = tdb_{X'}$.

PSTEP 5: Return the weighted frequent itemsets in the set of WF_X .

5 Experimental Evaluation

In this section, a series of experiments were conducted to compare the performance of the proposed projection-based weighted itemset mining algorithm (named *PWA*), and the traditional *WFIM* approach [6] with various parameter values. All the algorithms were implemented in J2SDK 1.6.0 and executed on a PC with a 2.53-GHz CPU and 4 GB of memory.

In the experiments, since it was difficult to obtain real datasets, a publicly available *IBM* data generator [4] was adopted to generate the required datasets. To simulate the real-world transaction database, a simulation model was derived from that used for utility mining [5] to generate the datasets. Note that the profits of the items were normalized to the weights ranging from 0.0 to 1.0 in the weight table according to the maximum profit among all items. In addition, the parameters used in the *IBM* data generator [4] were described as follows: T was the average length of items per transaction; I was the average length of maximal potentially frequent itemsets; N was the total number of different items; D was the total number of transactions.

Figure1 then showed the efficiency comparison of the two algorithms *PWA* and *WFIM* for the synthetic dataset under various minimum weighted support thresholds.

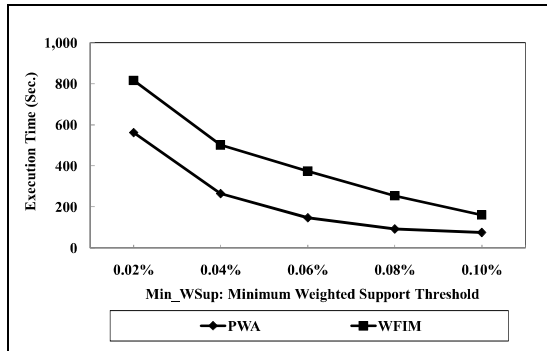


Figure 1. Efficiency comparison of the three algorithms on the T16I8N2KD200K dataset.

It could clearly be observed from the figures that the efficiency of the proposed *PWA* algorithm was better than that of the existing *WFIM* algorithm under different thresholds and data sizes. The main reason for this is that different from the existing model used in *WFIM*, the proposed *PWA* algorithm adopted the maximum weight in a transaction to keep the downward-closure property, and thus improving the proposed model to tighten more accurate upper-bounds of weighted supports for itemsets. Accordingly, the *PWA* algorithm outperformed the existing *WFIM* algorithms in finding weighted frequent itemsets.

6 Conclusions

This work presents an improved model named weighted support upper-bound (*WSUB*) to tighten upper-bounds of weight values of itemsets in mining. In addition, an efficient projection-based weighted itemset mining algorithm *PWA* is proposed to speed up the efficiency of finding

weighted frequent itemsets, and thus tightening more accurate upper-bounds of weighted support of itemsets. The experimental results show that the proposed *PWA* algorithm outperforms the *WFIM* algorithm in execution efficiency.

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